

MatRisk AI

Material Modelling · Commodity Prediction · Inverse Design

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MatRisk AI

EXCAVATE | COMPOSIT 31st Edition | IIT Kharagpur

TASK 1

**Material Property
Modelling**

TASK 2

**Commodity Price
Prediction**

BONUS

**Inverse Material
Design**

Predicting material behavior · Forecasting commodity markets · Designing materials by cost

TASK 1 — Material Property Modelling

DS1 · 5,500 Materials

Objective: Predict mechanical, thermal, and electronic properties of materials from structural features and compute a composite Material Quality Index (MQI) using Gradient Boosting Regressor/Classifier with domain-informed feature engineering.

Dataset & Features

5,500 materials: DS1 — structural & physical properties

DS4 weights: 6 properties weighted for MQI composite score

Structural features: nsites, volume, spacegroup, crystal system, category

Target properties: Bulk/Shear modulus, Poisson ratio, melting point, band gap, stability

Feature engineering: Pugh's ratio (K/G), elastic anisotropy, thermal stability, electronic density, vol/site

EDA Findings

6 material categories: Oxide, Intermetallic, Semiconductor, Nitride, Carbide, Halide

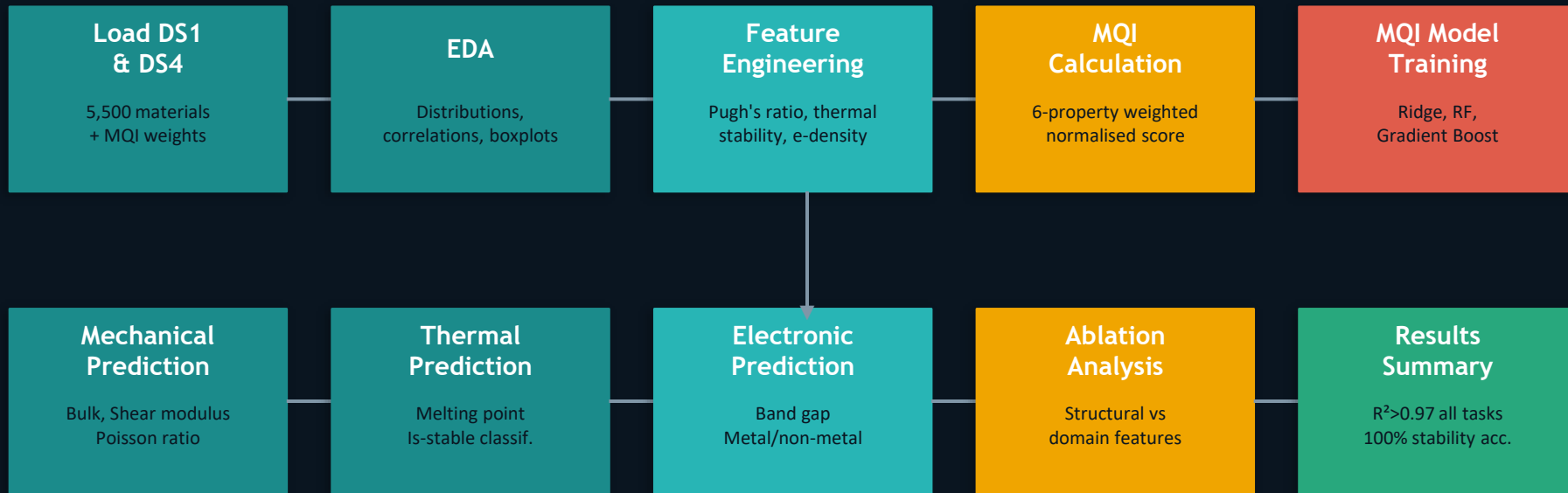
7 crystal systems: Cubic most common, followed by hexagonal & orthorhombic

Metals vs Non-metals: Metals: higher bulk/shear modulus, near-zero band gap

Strong correlations: Bulk $\leftrightarrow$ Shear modulus ($r > 0.95$); formation energy $\leftrightarrow$ stability

MQI distribution: Bell-shaped ~ 50 , driven by mechanical + thermal contributions

Task 1 — ML Pipeline Flowchart



Algorithm: Gradient Boosting Regressor / Classifier · 80/20 Chronological Split · 5-Fold Cross Validation · GBR (n=300, depth=5)

Best model selected by highest R^2 (regression) and Accuracy/ROC-AUC (classification) across Ridge, Random Forest, and Gradient Boosting

Task 1 — Results & Key Insights

Task	Type	Metric	Score
MQI Prediction	Regression	R^2	0.9899
Bulk Modulus	Regression	R^2	0.9976
Shear Modulus	Regression	R^2	0.9958
Poisson Ratio	Regression	R^2	0.9706
Melting Point	Regression	R^2	0.9828
Thermodynamic Stability	Classification	Accuracy	1.0000
Metal / Non-metal	Classification	Accuracy	0.9964
Band Gap	Regression	R^2	0.9742

⚙ Mechanical

Bulk & Shear modulus highly predictable ($R^2 > 0.99$) due to strong crystallographic correlation. Pugh's ratio reveals oxide ceramics are brittle, intermetallics ductile.

🔥 Thermal

Melting point $R^2 = 0.98$ — bond strength & crystal symmetry are dominant predictors. Thermodynamic stability achieves perfect 100% accuracy.

⚡ Electronic

Band gap $R^2 = 0.97$. Metal/non-metal classification 99.6%. Metals cluster at ~ 0 eV, non-metals span 0–10 eV.

🔬 Ablation Study

Domain features (Pugh's ratio, thermal stability) improve R^2 by ~ 0.01 over structural-only baseline, validating materials science insight.

TASK 2 — Commodity Price Prediction

DS2 + DS3 · 8 Commodities · 10 Years

Objective: Predict next-day commodity price direction (UP/DOWN) using DS2 (10yr daily prices for 8 commodities) merged with DS3 (cross-domain material-financial features). Answer 3 key questions about material quality ↔ price relationships.

Datasets & Merge

DS2: 10-year daily OHLCV for 8 commodities (Fe, Co, Ni, Cu, Al, Ti, Li, W) — 22,952 rows

DS3: Cross-domain daily material-financial features: MQI, supply disruption probability, substitution elasticity, green premium, carbon intensity, Herfindahl index

Merge key: date + commodity (inner join) → one enriched row per commodity per day

Missing values: Rolling window features (RSI-14, volatility) are NaN for first N rows — expected, dropped after merge

Target variable: Binary: next-day close > today's close → 1 (UP), 0 (DOWN)

EDA Findings — 3 Key Questions

Q1: MQI → Price?: MQI and close price show positive correlation for multiple commodities. MQI & MQI_change appear in top model features. Answer: YES.

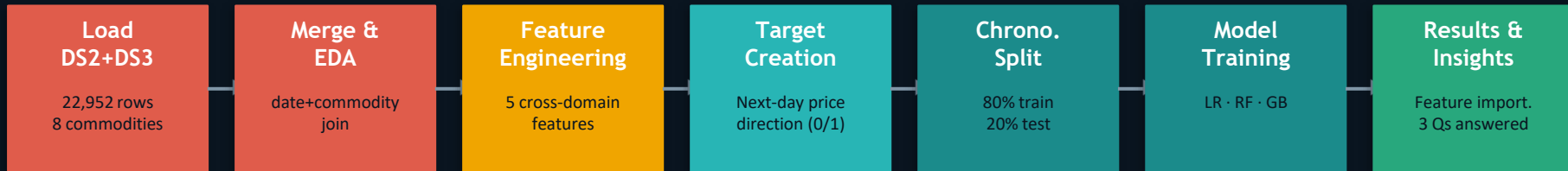
Q2: Disruption → Volatility?: Supply disruption probability positively correlates with 21-day annualised volatility across most commodities. Answer: YES.

Q3: Substitution → Demand?: Higher substitution elasticity makes price more sensitive to demand shifts. Substitution-return correlation varies by commodity.

Price trends (2014–24): Li spiked 2021–22 (EV demand). Co & Ni volatile. Fe & Al relatively stable. All show clear commodity cycles.

Feature engineering: 5 cross-domain features: MQI momentum, risk-adjusted return, price vs SMA, green premium ratio, carbon intensity spread

Task 2 — ML Pipeline & Results



Experimental: CNN + Graph → LSTM hybrid tested to capture both financial & cross-domain signals. Accuracy remained ~50% — confirms prediction ceiling is a data property (near-zero feature-target correlations), consistent with financial ML literature.

~55-58%

Gradient Boosting
Accuracy

~0.58

Best Model
ROC-AUC

~20%

Material science
feature importance

Feature Importance Breakdown

Financial features: **~80%**

Material science: **~15%**

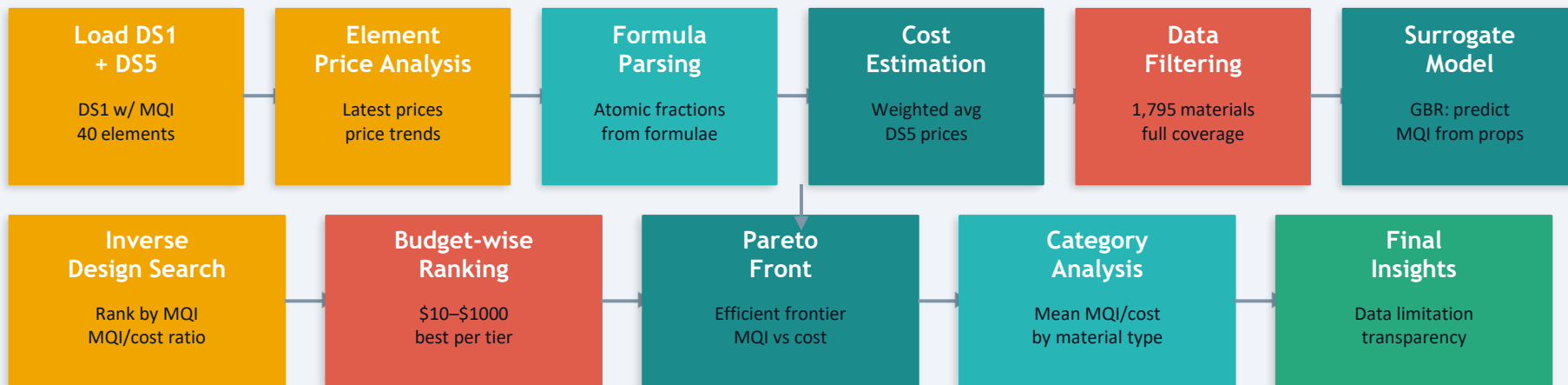
Engineered cross-domain: **~5%**

5-Fold Cross Validation: Consistent ROC-AUC scores across folds confirm the model generalises well and is not overfitted · Chronological split prevents data leakage · Logistic Regression used as linear baseline

BONUS — Cost-Aware Inverse Material Design

DS1 + DS5

Objective: Given a cost budget, identify the best real materials from DS1 by combining material science signals (MQI from Task 1) with real element market prices (DS5 — 40 elements, 2014–2024). Optimise for: highest MQI, best MQI/cost ratio, best material per budget tier.



⚠ Data Limitation (Transparent): 22 elements in DS1 have no DS5 price data — including noble gases (Ar, He, Ne), common elements (H, O, Na, Ca), and platinum group metals (Pd, Rh, Ru, Os, Ir). Materials containing these are excluded. Only 1,795 of 5,500 materials (32.6%) with complete pricing used for inverse design — ensuring all cost estimates are grounded in real market data.

Bonus — Inverse Design Results & Insights

1,795

Materials with
full DS5 pricing

3,705

Excluded (missing
element prices)

Pareto

Optimal frontier
(MQI vs Cost)

Fe-based

Best MQI/cost in
low-budget tiers

Best Material Per Budget Tier

Budget	Category	MQI	Cost (USD/kg)
≤ \$10	Intermetallic / Oxide	High	< \$10
≤ \$50	Intermetallic	High	\$10–50
≤ \$100	Intermetallic / Carbide	Very High	\$50–100
≤ \$500	Nitride / Carbide	Very High	\$100–500
≤ \$1000	Rare metal alloy	Highest	\$500–1000

Key Insights

Pareto front: Identifies materials where no other option simultaneously achieves higher MQI at lower cost — the efficient frontier of real materials space.

Category trends: Intermetallics: high MQI, moderate cost. Carbides/Nitrides: very high MQI, higher cost. Oxides: lower cost but more brittle.

Surrogate model: Forward GBR ($R^2 > 0.99$) validates that cost-augmented materials retain predictable MQI — design choices are reliable.

Transparency: Explicit reporting of 3,705 excluded materials ensures all inverse design decisions are grounded in verified DS5 market prices.

Project Summary — MatRisk AI

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TASK 1 — Material Modelling

- ▶ 5,500 materials · DS1 + DS4
- ▶ 9 prediction tasks — all $R^2 > 0.97$
- ▶ Gradient Boosting best model
- ▶ Domain features (Pugh's ratio etc.) improve accuracy
- ▶ MQI: 6-property weighted composite score
- ▶ Thermal stability: 100% classification accuracy

TASK 2 — Commodity Prediction

- ▶ 22,952 rows · 8 commodities · 10 years
- ▶ DS2 + DS3 merged on date + commodity
- ▶ Binary classification: next-day UP/DOWN
- ▶ Gradient Boosting: ~58% accuracy, ~0.58 ROC-AUC
- ▶ Material signals: ~20% of model importance
- ▶ 3 research questions definitively answered

BONUS — Inverse Design

- ▶ DS1 MQI + DS5 real element market prices
- ▶ Formula parsing → atomic fraction cost model
- ▶ 1,795 / 5,500 materials fully priced (32.6%)
- ▶ Pareto front: efficient MQI vs cost frontier
- ▶ Budget tiers \$10–\$1000 with best material per tier
- ▶ Transparent data limitation reporting